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ABSTRACT

Complex coacervation, a liquid–liquid phase separation phenomenon driven by electrostatic interactions between oppositely charged polyelectrolytes (PEs), has attracted widespread attention because of its relevance in biological systems and potential applications in materials science. Although many theoretical models, experimental investigations, and computational studies have investigated the thermodynamics, phase coexistence behavior, and rheological properties in great detail, a molecular-level understanding of the internal structure of the complex coacervate phase is still lacking. In this study, we investigate the effects of the degree of polymerization of the polyelectrolytes (N) on the phase behavior and internal structure of the resulting coacervate phase using molecular dynamics simulations employing a simplistic bead–spring model of polyelectrolytes and explicit nonpolar solvents. Our simulations show an increase in coacervate phase stability with N , elevating the critical temperature in agreement with existing theoretical predictions and experimental observations. The polyelectrolytes inside the dense phase maintain a homogeneous overlapping distribution without collapsing into globules. The compactness of the dense phase increases with N in agreement with prior experimental observations, despite a concomitant increase in the polymer's effective size as quantified by its radius of gyration (R_g). We discuss the implications of this model for a fundamental understanding of the coacervation process and as a first step toward the systematic examination of the mutual role of electrostatics and chemistry in the behavior of solvated polyions.

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I. INTRODUCTION

Polyelectrolyte complexes (PECs) are an interesting class of materials formed through the associative phase separation of oppositely charged polyelectrolytes in aqueous solution.^{1–5} This process, primarily driven by electrostatic interactions, results in the formation of dense polymer-rich phases that can manifest as liquid-like coacervates or solid-like precipitates.^{5–10} Although solid-like glassy precipitates have only recently garnered attention for applications in multilayer films, functional coatings, and biomimetic substrates,^{11–13} liquid-like coacervates have been of interest since the seminal work of de Jong and Kruyt¹⁴ in the early twentieth century. Complex coacervates have emerged as critical components

in both biological and synthetic systems. In nature, biomolecular condensates formed via liquid–liquid phase separation (LLPS) play essential roles in cellular organization and processes. For example, DNA–histone interactions that compact DNA into chromatin fibers,¹⁵ adhesive secretions in sandcastle worms,¹⁶ and compartmentalized nucleotide organization in bacteria¹⁷ are all thought to involve phase-separated condensates. Coacervates have also been conceptualized as model protocells and are hypothesized to have played a potential role in the origin of life.^{18–20} Furthermore, due to their dynamic fluid-like behavior and ability to encapsulate biomolecules, coacervates have attracted considerable attention in biomedical and biotechnological applications. They have been engineered as drug delivery vehicles,^{21–23} bioadhesives,²⁴ and tissue

scaffolds.²⁵ In addition, coacervates have been employed as model artificial bioreactors,²⁶ encapsulating mediums for biomolecules,^{27,28} and enzyme activity enhancers.^{29–31} Beyond equilibrium structures with liquid- or solid-like properties, nonequilibrium assemblies, such as layer-by-layer arrangements of oppositely charged polymers and multilayered polymer capsules, have demonstrated utility in diverse technological applications. These include active coatings,^{32,33} functional membranes,^{34–36} drug delivery systems,^{37,38} and biomimetic substrates.²⁵

The physicochemical properties of PECs are inherently tunable and depend sensitively on a range of parameters, including polymer molecular weight,^{39,40} charge density,^{41,42} the mixing ratio of oppositely charged polymers,^{42,43} ionic strengths and valencies,^{8,44–46} pH,^{47–49} and solvent quality.^{50,51} Modulating these variables alters the phase behavior, viscoelastic properties, internal structure, and thermodynamic stability of the resulting coacervate or complex. Understanding how these parameters influence coacervate behavior has prompted the development of theoretical models to capture the underlying thermodynamics and molecular interactions governing complex coacervation. Theoretical models for polyelectrolyte complexation range from the simple Voorn–Overbeek model,⁵² which uses basic thermodynamic and electrostatic principles, to more complex approaches such as Random Phase Approximation (RPA)^{53,54} and field-theoretic simulations^{55,56} that incorporate charge fluctuations and connectivity. As an alternative method, the polymer reference interaction site model (PRISM)-based liquid-state theory^{57,58} focuses on the correlations between chains and counterions, also accounting for the excluded volume. These models provide predictions for phase behavior, critical salt concentrations, and counterion partitioning, with varying degrees of agreement with experimental data.³ Although these theoretical predictions have some degree of disagreement regarding the order of phase transition during complexation and ion concentration within the complex phase, the prediction that an increase in the size of the polyelectrolytes enhances complexation, leading to a wider two-phase region and an increased critical point, remains unchanged. In addition, simulation techniques have also been employed to better understand the molecular picture of PE complexation.^{59–70} Atomistic and coarse-grained molecular dynamics simulation studies have reported the effects of polyelectrolyte size and salt concentration on coacervate size, density, and compactness. In addition, experimental studies^{71–75} have provided notable insights into the internal structure of polyelectrolyte coacervates. Several reviews have summarized these developments and are recommended for further context.^{7,76–80}

Typically, coarse-grained models that employ implicit solvents are utilized for exploring coacervation, provide significant computational advantages, and excel at capturing the theoretically predicted and experimentally observed behaviors. This includes the qualitative dependencies of phase diagrams on the degree of polymerization (N), the scaling of radius of gyration (R_g) with N , and the ion/salt partitioning behavior. However, these models fail to capture the critical entropic contributions from solvent structuring and expulsion, a process that governs coacervate formation and stability. Consequently, by neglecting solvent-derived excluded volume effects, they also offer very limited insight into the internal structure of coacervates, a factor essential for optimizing these materials for biomedical and industrial applications. Although some MD simulation studies^{81–86} have implemented explicit solvents to understand

the effects of the ionic environment, chirality, and hydrogen-bonded interactions on the coacervation of specific systems, those are of a single type of condensate, focusing on a single state point of the phase diagram, and do not systematically investigate the phase behavior of two-phase PE solutions and the internal structure of the dense phase. This is an important concept that underlies the functionality and stability of polyelectrolyte solutions of different types. In the present study, we focus on the following questions: (I) Does our simplified model, which incorporates a nonpolar explicit solvent, adequately capture the qualitative relationship between the critical point and the degree of polymerization of polyelectrolytes (PEs)? (II) Can this model be used to investigate the internal structure of the dense coacervate phase and its qualitative dependence on the degree of polymerization of polyelectrolytes? The implementation of explicit solvent beads in our model to trace out the whole phase diagram of the polyelectrolyte system (1) enables structural comparisons when solvent is present and (2) creates a baseline for studying mixed solvent conditions and/or crowding agents in the solvent in a more realistic way, enabling modeling of intracellular biomolecular condensates. To our knowledge, this is one of the first simulation studies that could potentially connect with measurements regarding polyelectrolyte condensate structure under conditions where solvent may access the interstitial voids of the condensate, thus providing a more realistic physical picture.

II. METHODS

Our systems comprise polyanions and polycations in the presence of an explicit solvent. Polyelectrolytes are modeled following the Kremer–Grest coarse-grained representation of polymers in solution.⁸⁷ Each of our systems contains 48 000 total beads, including 2000 beads each of polycations and polyanions, within a simulation box of dimensions $23.2\sigma \times 23.2\sigma \times 139.2\sigma$, where σ is the diameter of each PE and solvent bead. This results in a constant mole fraction of polyelectrolyte beads ($\approx 9\%$) and a fixed number density of beads $\rho = 0.64/\sigma^3$, maintaining the liquid state of the Lennard-Jones beads.⁸⁸ We vary the polymer chain lengths (N) to 10-mer, 20-mer, 30-mer, and 40-mer, with polycation and polyanion chains maintained at equal lengths within each system. We keep the total concentration of polymer beads constant in all systems, except for the 30-mer case, where the total number of polymer beads is 3960, compared to 4000 in the other systems. To our polyion chains, we employ a FENE bond potential [Eq. (1)] for bonded interactions with bond constant $K = 100k_B T/\sigma^2$ and a maximum bond length $R_0 = 2.0\sigma$,

$$E_{\text{FENE}}(r) = -\frac{1}{2}KR_0^2 \log \left[1 - \left(\frac{r}{R_0} \right)^2 \right], \quad (1)$$

where r denotes the distance between two particles. For the non-bonded interactions, we use a shifted-truncated Lennard-Jones potential [Eq. (2)] with a cutoff radius of $2^{1/6}\sigma$ for van der Waals (vdW) interactions,

$$E_{\text{LJ}}(r) = 4\epsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right], \quad (2)$$

$$E_{\text{STLJ}}(r) = \begin{cases} E_{\text{LJ}}(r) - E_{\text{LJ}}(r_c), & r < r_c, \\ 0, & r \geq r_c, \end{cases}$$

and a long-range full Coulombic potential function [Eq. (3)] with a cutoff radius of 5σ to account for electrostatic interactions between charged polymer beads,

$$E_{\text{coul}}(r) = \frac{kq_1^* q_2^*}{\epsilon r}, \quad (3)$$

where k is an energy-conversion constant, q_1^* and q_2^* are the reduced charges of the PE beads, and ϵ is the dielectric constant of the medium. The cutoff distances for the interactions between the solvent and polymers are 2.5σ and 5σ for the vdW and Coulomb potentials, respectively. To calculate long-range Coulomb interactions, a particle–particle particle–mesh (pppm) solver⁸⁹ is used.

The dimensions of the simulation box and the initial configurations are defined using the software package *moltemplate*,⁹⁰ in which polymer chains are placed in the center of the simulation box and surrounded by solvent beads. For all polymeric systems, initial equilibration involves a series of runs in which the time step is gradually increased to $10^{-3}\tau_{\text{lj}}$. Following a system energy minimization, this process was performed at $q^* = 3.0$ and $T^* = 2.0$ for 10×10^6 time steps using the *NVT* ensemble to ensure a slab-like arrangement of polyelectrolytes at the box center and uniform solvent dispersion throughout. Following initial equilibration, each system then undergoes a charge-sweep protocol, during which the effective charge of the monomers (q^*) is varied from 0.50 to 1.50 in increments of 0.10. The equilibration was carried out in the *NVT* ensemble at $T^* = 1.0$ while utilizing a Langevin thermostat.^{91,92} By modifying q^* , the Coulomb coupling constant Γ can be altered, allowing a systematic modification of the strength of interaction between polyelectrolyte chains. The reduced temperature and reduced charges are related through the dimensionless Coulomb coupling constant Γ , as described in the following equation:

$$\Gamma = \frac{\lambda_B}{\sigma} = \frac{q_1^* q_2^*}{\epsilon T^*}. \quad (4)$$

Here, T^* is the reduced temperature of the system, λ_B is the effective Bjerrum length, q_1^* and q_2^* are the reduced charges of the PE monomers comprising the polyelectrolyte chains, and ϵ is the dielectric constant of the medium. We note that typical hydrated

monovalent ions exhibit a weak Coulomb coupling constant of $\Gamma \approx 1.0$, although solutions of strong electrolytes such as polystyrene sulfonate samples (PSS) are often modeled by their contact Γ value of ≈ 2.6 .^{93,94}

Increasing the polymer charge (q^*) in our model effectively lowers the system temperature in the analogous solvent-free models, as the enhanced electrostatic adhesion between polyelectrolytes drives complex coacervate and precipitate formation, with solid aggregates observed at high values of q^* (e.g., $\gg 1.50$). This protocol identifies charge values that support biphasic liquid–liquid coexistence, a state hypothesized to mimic conditions compatible with cell-like structural integrity. This protocol can also serve as a salt concentration analog in the same way temperature–salt effects are often discussed. By modulating the polyelectrolyte charge, the effective interchain interactions are adjusted without direct salt addition. This mimics the enthalpic screening of ions/salts (e.g., the reduced electrostatic repulsion). In short, a reduction in the polymer bead charge (q^*) corresponds to an effective increase in the salt concentration or temperature of the system. Figure 1(a) presents a schematic phase diagram in the salt concentration/temperature and polymer concentration plane (upper panel) and how a hypothetical phase diagram is expected to appear in q^* and polymer concentration plane (lower panel).

At each q^* value, the system undergoes additional equilibration for at least 30×10^6 time steps. To ensure proper equilibration, we monitor these systems by dividing the simulation box into quadrants and measuring the solvent density difference between the inner and outer regions. A difference that begins to fluctuate around a constant value indicates that the system has reached a steady state. A visualization of this process is presented in Fig. S1 (see the [supplementary material](#)). The effect of q^* on a randomly chosen polymeric system is depicted in Fig. 1(b) as a series of snapshots of well-equilibrated simulation boxes. For each system, a production run of 50×10^6 time steps is performed in the *NVT* ensemble, using a time step of $10^{-3}\tau_{\text{lj}}$ and saving the trajectories every 10^5 steps. All simulations are performed using the MD simulation package LAMMPS,^{95–102} version 7Feb2024–update 1, employing standard dimensionless Lennard–Jones (LJ) reduced units. Throughout this study, we adopt PE bead mass $m_{\text{bead}} = 1$, $\sigma = 1.0$, $\epsilon = 1.0$, energy-conversion constant for

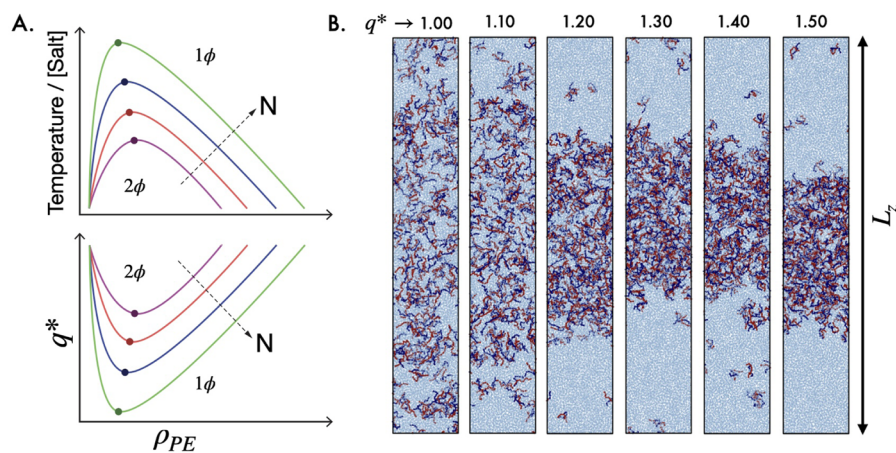


FIG. 1. (A) Schematic phase diagram of polyelectrolyte systems in the plane formed by temperature/salt concentration and polyelectrolyte bead concentration (ρ_{PE}) (upper panel) and hypothesized phase diagram in the polyelectrolyte bead charge (q^*) and ρ_{PE} plane (lower panel). (B) Visual representation of one representative polyelectrolyte (PE) system (10-mer PE solutions) for varying charges of the PE beads. The charge of the PE beads determines the transition from a single-phase system to a two-phase system. Polycations are shown in red, while polyanions and solvent beads are depicted in dark blue and light blue, respectively.

the Coulombic potential $k = 1.0$, $\tau_{ij} = 1.0$, and dielectric constant $\epsilon_r = 1.0$, as these values are typical for LJ units.

Our key analytical metrics include (I) phase diagram, (II) PE radius of gyration statistics, and (III) Voronoi tessellation analysis statistics. While performing the Voronoi tessellation analysis, only PE beads are considered as site points. For each pair of sites, the midpoint of their connecting line segment is calculated, and a perpendicular plane is constructed at that midpoint. These perpendicular bisectors serve as boundaries, and their intersections define the edges of the Voronoi cells. The cavity radius (r_{cavity}) is defined as the distance from a site to the farthest vortex of its Voronoi cell, which can offer a quantitative measure of the mesh size of the polyelectrolyte matrix within the dense coacervate phase. This metric is often measured experimentally using scattering techniques to characterize the internal structure of the dense coacervate phase. A higher average cavity radius may be associated with a less compact polyelectrolyte coacervate phase in this context. All analyses were carried out utilizing the OVITO¹⁰³ Pro Python module.

These results collectively reveal how the chain structure and phase behavior evolve with the degree of polymerization (N). Prior to analysis, all trajectories were remapped and rewrapped to align the center of the polyelectrolyte complex phase with the center of the simulation box. Throughout this study, we restrict our analysis to values of q^* that yield a stable, well-equilibrated two-phase system. Systems exhibiting instability in the two-phase boundary (e.g., $q^* \leq 1.10$ for 10-mer polymeric systems) are excluded from further analysis.

III. RESULTS AND DISCUSSION

To elucidate the rationale for employing the computationally expensive explicit solvent model, we simulated two $N = 40$ PE systems, one with and one without explicit solvent. In both cases, identical run parameters, as detailed in Sec. II, were applied. When the densities of the coacervate phase are compared (see Fig. 2), we observe that the coacervate simulated in the absence of explicit solvent beads exhibits a density that is approximately twice that observed in simulations incorporating explicit solvent. This suggests that the inclusion of explicit solvent beads is necessary to get a realistic internal structure of the coacervates. Although the effect of solvent excluded volume can be incorporated using effective repulsions or osmotic pressure derived from (e.g.) Carnahan–Starling terms,^{104–107} explicitly including solvent particles provides a more straightforward and realistic representation. It should be noted that, throughout this study, the solvent beads are modeled with the same size as the “solvated” polymer beads, a choice that may influence the calculated potential of mean force (PMF). Realistic solvent molecules are much smaller than the monomer units of polymers represented as polymer beads in this coarse-grained description. This can lead to an underestimation of the solvent packing inside the polymer matrix and misrepresentation of the solvent entropy and free energy profiles.¹⁰⁸ As the focus of this study is to qualitatively understand how the phase diagram and internal structure of coacervates depend on the degree of polymerization of the polyelectrolyte chains, our simplified model employing explicit solvents can represent a valuable first step in this direction.

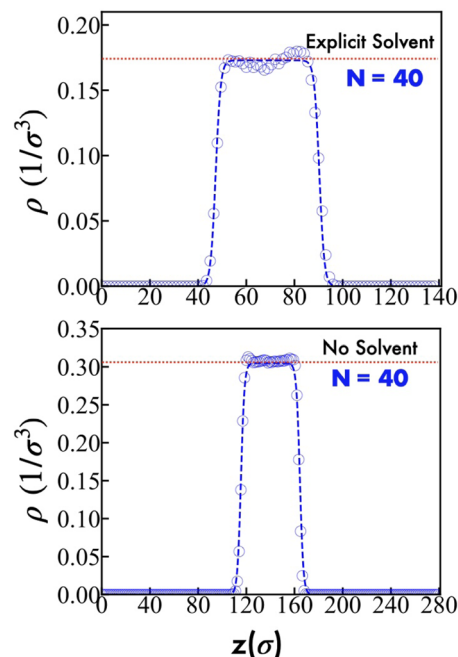


FIG. 2. Density profile of polyelectrolyte (PE) beads [simulated (open symbols) vs fitted (dashed line)] for $N = 40$. The upper panel is simulated with explicit solvent beads, and the lower panel is simulated without solvent beads. The coacervate simulated in the absence of explicit solvent beads exhibits a density that is approximately twice that observed in simulations incorporating explicit solvent.

A. Phase behavior

The phase behavior of our systems is analyzed by first extracting the capillary geometry for both the low-density (supernatant) and high-density (coacervate) phases. From this, the corresponding PE density profiles and extracted PE density in both coacervate (polymer-dense) and supernatant (polymer-dilute) phases are calculated following the fit function adopted from Ref. 109:

$$\rho(z) = \rho_S + \frac{1}{2}(\rho_C - \rho_S) \left[\tanh \frac{(z - z_{0L})}{d} + \tanh \frac{(z_{0R} - z)}{d} \right], \quad (5)$$

where ρ_C and ρ_S are the PE densities in the coacervate and supernatant, respectively; z_{0L} and z_{0R} are the locations of Gibbs dividing surfaces (GDSs) of the interface at the left and right sides of the coacervate profile, respectively; and $2d$ is the thickness of the GDS. ρ_C , ρ_S , z_{0L} , z_{0R} , and d are the adjustable parameters in the nonlinear fit of the obtained profile. We note that the first hyperbolic tangent function within the second term of Eq. (5) fits the left surface of the coacervate [Fig. 3(a)], while the second describes the right surface. This fitting function [Eq. (5)] offers a more precise estimation of coacervate and supernatant densities because it does not inherently assume a symmetric PE density profile about the center of the profile. A simulated PE density profile along with the fitted one for a randomly selected system is visualized in Fig. 3(a). The phase diagram is constructed in the space spanned by the reduced PE bead charge (q^*) and the PE bead concentration ρ_{PE} for all polymeric systems with varying

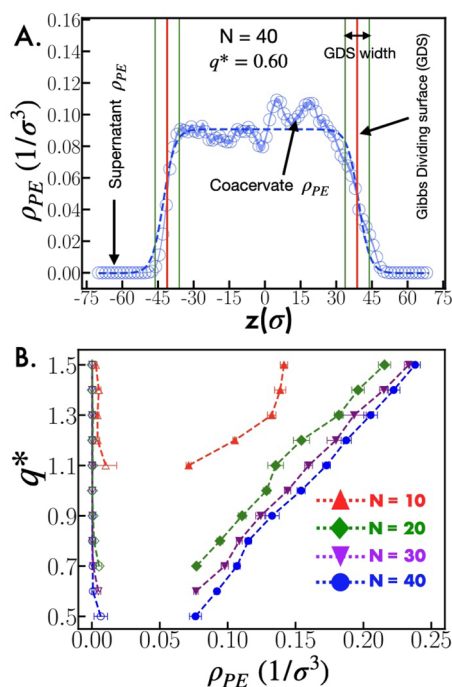


FIG. 3. Structural and phase behavior analysis of polyelectrolyte systems. (A) Density profile of polyelectrolyte (PE) beads (simulated vs fitted) for a representative system ($N = 40$, $q^* = 0.60$). (B) Unified phase diagram in the (q^*, ρ_{PE}) parameter space, showing coacervate and supernatant phases for all polymeric systems with varying N .

degrees of polymerization (N). The simulated phase diagram of the systems with varying N is presented in Fig. 3(b).

The phase boundary (onset of coacervation) shifts to lower q^* values as N increases. This allows longer polyelectrolyte chains (e.g., 40-mer) to phase separate at lower PE bead charges than shorter chains (e.g., 10-mer). The coacervate phase becomes more concentrated in polymers as N increases. A qualitatively similar type of phase diagram in the $q^* - \rho_{PE}$ parameter space has been reported earlier.¹¹⁰ In our simulations with fully charged, moderately long chains ($N \geq 10$), the concentration of PEs in the supernatant is negligible under conditions away from the critical point. Even with this low PE concentration in the supernatant, the trend that the dilute phase becomes more polymer-dilute for longer chain polymeric systems is observed. This can be clearly visualized from the supernatant densities near the critical points of any particular system (e.g., $q^* = 1.10$). Essentially, as N increases, the phase envelope broadens and the coacervate phase becomes more stable, enabling phase separation at lower q^* or higher temperatures. Furthermore, our simulations show that the degree of densification of the coacervate phase decreases as N increases. This effect is most pronounced when increasing the chain length from $N = 10$ to $N = 20$. The change in coacervate density is less significant between $N = 20$ and $N = 30$ and becomes even smaller between $N = 30$ and $N = 40$. These findings are qualitatively consistent with classical theories (e.g., Voorn-Overbeek⁵² and scaling arguments¹¹¹), as well as theoretical¹¹⁰ and experimental observations.^{64,112,113}

B. Composition of the coacervate phase

Figure 4(a) shows the density profiles of the polyelectrolyte and solvent beads for a specific q^* across all polymeric systems. The simulated dense phase is observed to be a homogeneous mixture of solvent and polyelectrolytes, in agreement with experimental observations.^{71,112} The density of the PEs in the coacervate phase increases with N at a fixed interaction strength ($q^* = 1.10$), driven by the greater reduction in conformational entropy accompanying chain elongation. In particular, while four pairs of 10-mers, two pairs of 20-mers, and one pair of 40-mers have the same number of beads and can achieve comparable enthalpies in terms of interaction sites, the “cut” bonds in the 20-mer and 10-mer cases allow an increasing amount of translational entropy for those polymers. Since the 40-mer suppresses entropy while simultaneously increasing the probability of contact formation, enthalpic interactions eventually dominate, resulting in a denser polyelectrolyte complex (PEC). Figure 4(b) shows the solvent content (wt. % of solvent beads) inside the dense coacervate as a function of N at $q^* = 1.10$. Our model predicts a solvent content of 55%–70% within the coacervate phase. For the 10-mer PE system, solvent content is $\approx 70\%$, and it decreases systematically with increasing chain length (N) and reaches $\approx 55\%$, indicating progressive solvent release followed by expulsion from the coacervate phase as polymer chains become longer.

Existing experimental observations^{41,62,112,114,115} usually report the water content within the coacervates far from the critical point

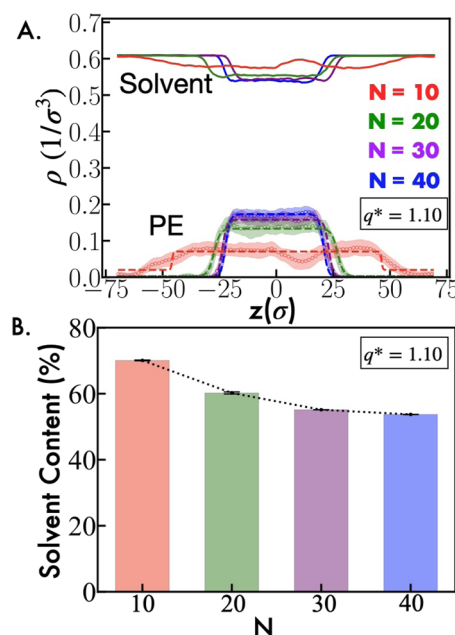


FIG. 4. Structural and compositional analysis of coacervate phases. (A) Simulated (open symbols with associated error) and fitted (dashed lines) density profiles of polyelectrolytes (PEs) and simulated solvent bead density profiles (solid lines) across polymeric systems with varying chain lengths (N) at a fixed interaction strength ($q^* = 1.10$). (B) Solvent mass fraction within the coacervate phase (including Gibbs dividing surfaces) as a function of chain length (N) at the same interaction strength ($q^* = 1.10$).

over a wide range, roughly between 50% and 85%, depending on the charge density of the PEs, pH, and salt concentration. The complex coacervates with fully charged polyelectrolytes in the absence of salts are known^{41,62} to have water content within 50%–55%. Our estimated solvent content for longer chain polymeric systems qualitatively agrees with these experiments.

Furthermore, in prior experimental work, Spruijt *et al.*¹¹² reported that the solvent content of salt-free polyelectrolyte coacervates becomes effectively independent of polymer chain length (N) once $N \geq 200$, whereas below this threshold, it exhibits a weak, decreasing dependence on N . Similar indirect observations were reported⁶⁴ in the context of variation of water content within the coacervate phase with coacervate polymer concentration. Our coarse-grained simulation model qualitatively mirrors this behavior. The change in the coacervate solvent content between $N = 30$ and $N = 40$ is negligible, demonstrating the model's capacity to reproduce the plateauing of the solvent content in the long-chain limit.

C. Polyelectrolyte conformation inside the complex coacervate

The probability distribution of the radius of gyration (R_g) of the PEs is presented in Fig. 5(a) for a representative q^* . The distribution of R_g shifts progressively toward higher values and becomes broader as the polymer chain length (N) increases. The R_g distributions for the 20-mer, 30-mer, and 40-mer systems exhibit substantial overlap, while the distribution for the 10-mer is distinctly separate, showing minimal overlap with those of longer chains. This narrow R_g distribution for the 10-mer polyelectrolytes reflects their limited capacity to form extensive polymer networks. In contrast, longer chains promote the formation of a more interconnected network of polyelectrolyte chains, resulting in a denser coacervate phase. To evaluate the solvent quality relevant to polyelectrolytes in the complex coacervate phase, the scaling behavior of the radius of gyration with N is tested, following the relationship $\langle R_g \rangle \sim N^\nu$, where ν is the Flory exponent. Here, $\langle R_g \rangle$ is the mean value of corresponding

R_g distributions. In this scaling relationship, an exponent of $\nu = 0.5$ signifies a theta solvent, while $\nu > 0.5$ and $\nu < 0.5$ correspond to the good and poor solvents, respectively. This scaling behavior is illustrated in Fig. 5(b) for two representative q^* values. Our estimated Flory exponent values (ν) are observed to be consistently ≈ 0.70 across all simulated systems. These results suggest that the dense coacervate phase is a homogeneous liquid with polyelectrolytes having an expanded coil state conformation. These simulated PE chains are more spaced out than the “good solvent” limit and hint at some elongation and rigidity imposed by the backbone charges. We anticipate that this effect will be less prominent and tend toward the $\nu \approx 0.6$ limit as $N \rightarrow \infty$. Note that experiments^{71,72,112} have reported Gaussian-like conformations ($\nu \approx 0.5$) of the polyelectrolyte chains within the coacervate, which might be facilitated by the presence of a polar solvent (water) and a nominal amount of salt in dilute salt conditions. This can effectively screen the intramolecular repulsion between polyelectrolyte beads, leading to slightly more compact chains.

Next, we explore the effect of varying q^* on the PE chain conformation, as characterized by R_g . Figures 5(c) and 5(d) illustrate the distribution of $\langle R_g \rangle$ of PE chains at varying q^* values ranging from $q^* = 1.10$ to 1.50 for two randomly selected polymeric systems, $N = 10$ and 40, respectively. The change in R_g distribution for all the polymeric systems with q^* is minimal in our examined range of q^* . This agrees with the experimental findings,^{71,74,116} which report a very weak dependence of polyelectrolyte chain conformations on salt concentration or temperature within the dense coacervate phase. The $\langle R_g \rangle$ values for tested charge values across all polymeric systems are provided in Tables S1–S4 (see the [supplementary material](#)).

D. Compactness of the dense phase: Voronoi tessellation analysis

To further investigate the effect of N on the internal structure of the dense coacervate phase, the probability distribution of cavity radii [$P(r_{\text{cavity}})$] and the average cavity radius, $\langle r_{\text{cavity}} \rangle$, across all

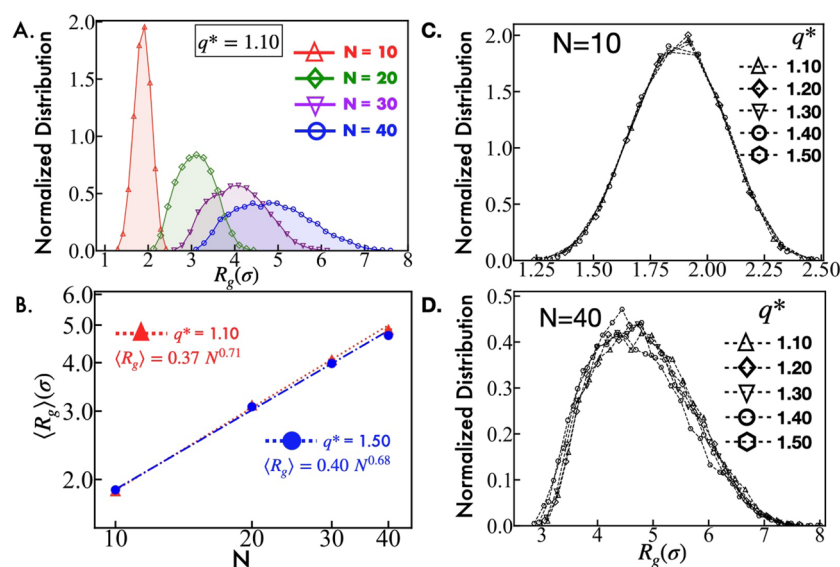


FIG. 5. Chain conformation analysis in coacervate phases. (A) Radius of gyration (R_g) distributions for polyelectrolytes (PEs) with varying chain lengths (N) at a fixed interaction strength ($q^* = 1.10$). (B) Scaling relationship between $\langle R_g \rangle$ and N at two interaction strengths ($q^* = 1.10$ and 1.50). The respective scaling equations are provided as insets within the graph. (C) Distribution of R_g at varying q^* for $N = 10$. (D) Distribution of R_g at varying q^* for $N = 40$.

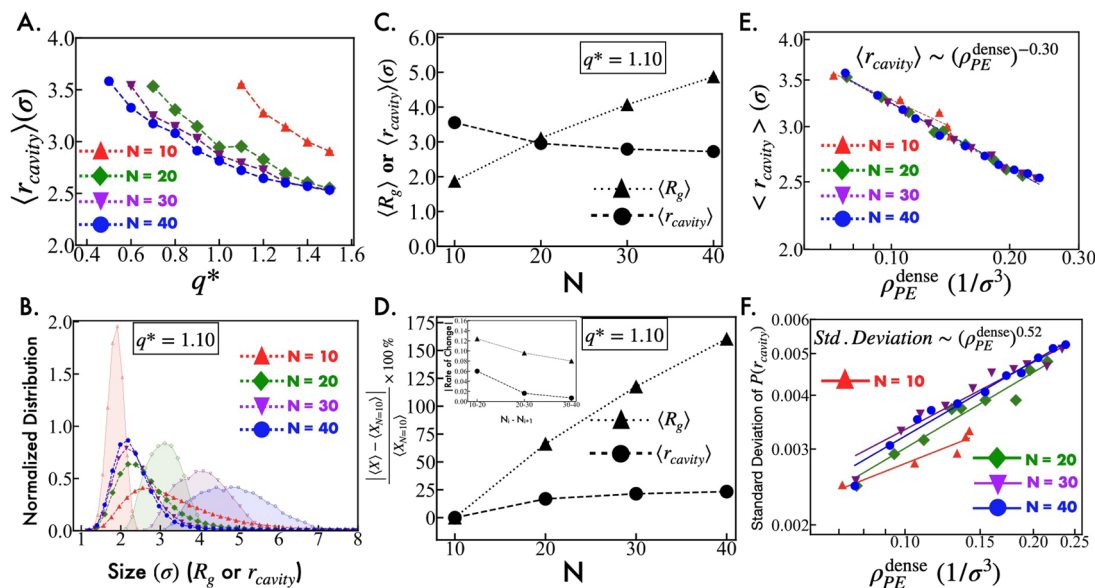


FIG. 6. Effective PE chain size and cavity size analysis within the coacervate phase. (A) Average Voronoi cavity radius, $\langle r_{\text{cavity}} \rangle$, as a function of charge q^* across all polymeric systems of varying N . (B) Distributions of the PE radius of gyration (R_g) (solid line with open symbol) and PE matrix cavity radius (r_{cavity}) (dashed line with closed symbol) at a fixed interaction strength ($q^* = 1.10$) across systems with varying chain lengths (N). (C) Mean radius of gyration ($\langle R_g \rangle$) and Voronoi cavity radius ($\langle r_{\text{cavity}} \rangle$) as functions of degree of polymerization (N) at $q^* = 1.10$. (D) Absolute relative change (in %) of $\langle R_g \rangle$ and $\langle r_{\text{cavity}} \rangle$ compared to their 10-mer PE system counterpart. The inset shows the absolute rate of change of these parameters as N changes from 10 to 20, 20 to 30, and 30 to 40. (E) Mean Voronoi cavity radius, $\langle r_{\text{cavity}} \rangle$, as a function of PE bead density within the coacervate phase ($\rho_{\text{PE}}^{\text{dense}}$) across all polymeric systems of varying N . The corresponding scaling relationship is shown as an inset within the figure. (F) Standard deviation of the distribution of cavity radii as a function of $\rho_{\text{PE}}^{\text{dense}}$ across all polymeric systems with varying N . The corresponding scaling relationship is shown as an inset within the figure.

simulated polymeric systems are calculated. Figure 6(a) presents the average cavity radius, $\langle r_{\text{cavity}} \rangle$, as a function of q^* across all polymeric systems with varying N . The $\langle r_{\text{cavity}} \rangle$ values for all tested charge values across all polymeric systems are provided in Tables S1–S4 (see the [supplementary material](#)). The cavity size within the polyelectrolyte network in the coacervate decreases with both increasing q^* and chain length (N) across all systems. This reduction is most pronounced when transitioning from 10-mer to 20-mer chains, whereas the differences between the 20-mer to 30-mer and 30-mer to 40-mer systems diminish progressively, aligning with the coacervate density trend [Fig. 3(b)]. This implies that the coacervate phase becomes increasingly compact as the interaction strength between the PEs increases. This result is intuitive and qualitatively agrees with an experimental observation⁷¹ in which the mesh size of the complex coacervate formed by a solution of poly(acrylic acid) and poly(N,N-dimethylaminoethylmethacrylate), measured using neutron and x-ray scattering, decreased with decreasing salt concentration, that is, with increasing effective interaction strength (q^*) between PEs. Next, we examine whether a correlation exists between the effective size of the individual polyelectrolytes and the compactness of the coacervate phase. Figure 6(b) shows the distribution of R_g and r_{cavity} across all systems for a representative q^* value ($q^* = 1.10$). With increasing N , the peak of the cavity radius distribution progressively shifts toward smaller cavity sizes. This indicates that the PE matrix within the coacervate becomes increasingly compact with increasing N . Although the cavity radius distributions of longer-chain PE

coacervates are qualitatively similar, the distribution for the 10-mer PE coacervates is distinctly different. For $N = 10$, the peak appears at a larger cavity radius value, and the distribution becomes considerably broader compared to the longer-chain PE systems with $N \geq 20$. The significantly less compact structure of the 10-mer coacervate results from the balance between two competing forces: (I) electrostatic inter-chain enthalpic associations and (II) thermal fluctuations. In summary, shorter chains form weaker enthalpic associations and are more vulnerable to thermal fluctuations, making the 10-mer coacervate significantly less compact and dense than coacervates formed by longer polyelectrolyte chains. When the cavity radii distributions are compared with the distributions of R_g of the individual PEs of the respective systems [see Fig. 6(b)], we find that even though the R_g distribution peaks shift to the higher R_g values by a considerable amount as chain length elongates, the decrease in cavity radii peaks does not change significantly for $N \geq 20$. Figure 6(c) illustrates $\langle R_g \rangle$ and $\langle r_{\text{cavity}} \rangle$ as a function of N at the same q^* . Then, the absolute relative change in those average quantities is calculated and compared to that of the 10-mer PE system [Fig. 6(d)]. This clearly shows that the change in the effective size of the individual PEs does not correlate with their Voronoi cavity size. The inset of Fig. 6(d) shows the absolute rate of change of these parameters with N , which clearly establishes that the mesh size distribution within the coacervate phase is largely independent of the constituent polyelectrolyte size, especially for longer-chain PEs ($N \geq 20$). This observation is intuitive, correlates with prior density trends, and has

been tested experimentally.⁷¹ Next, we discuss the possible charge-shape superposition observed in our simulated systems. In Fig. 3(b), we observe that the coacervate polyelectrolyte density remains similar across all values of q^* regardless of chain length. A similar trend is found in Fig. 6(a) with the average cavity sizes. How, then, does $\langle r_{\text{cavity}} \rangle$ scale with the packing density of PE beads within the coacervate phase, $(\rho_{\text{PE}}^{\text{dense}})$? By fitting all polymeric systems via a scaling law, $\langle r_{\text{cavity}} \rangle \sim (\rho_{\text{PE}}^{\text{dense}})^{-\xi}$, the average scaling factor (ξ) is calculated and presented in Fig. 6(e). All the polymeric systems show a universal scaling law with $\xi \approx -1/3$. This is anticipated, as by definition, the average volume of a Voronoi cell ($\langle V \rangle$) is inversely proportional to the packing density (C) of beads with diameter d , as given by the relationship $\langle V \rangle = \frac{1}{6} \frac{\pi d^3}{C}$.¹¹⁷ It would be of interest to explore how the inclusion of polar solvents and ions might influence this behavior in future studies. To explore any possible correlation between the cavity radii distribution spread with the density of PE beads within the coacervate phase of all tested N , the scaling relationship of standard deviation of $P(r_{\text{cavity}})$ with $\rho_{\text{PE}}^{\text{dense}}$ is estimated with a scaling equation, $\text{Std. Deviation} \sim (\rho_{\text{PE}}^{\text{dense}})^{\chi}$, to estimate the scaling factor χ [Fig. 6(f)]. A near-universal scaling law is observed here for all the polymeric systems, with an average value of $\chi \approx 0.5$. The scaling parameter values for both cases are listed in Table S5 (see the [supplementary material](#)). Although the origin of this scaling law remains unclear, our simulations reveal a potentially useful correlation between the cavity distribution within the coacervate phase and polyelectrolyte densities.

IV. CONCLUDING REMARKS

In this work, we explore a novel model to elucidate the internal structure of polyelectrolyte coacervates with our systems involving explicit solvent and oppositely charged polyelectrolyte chains with varying degrees of polymerization. Through brute-force molecular dynamics simulations, we systematically vary the polyelectrolyte bead charges to mimic the effects of factors such as salt concentration, temperature, or a combination of these influences. The phase behavior exhibited by our systems was consistent with experimental observations and theoretical predictions. In particular, the stability of the coacervate phase increases with the degree of polymerization of the PEs. In addition, our results show that the radius of gyration (R_g) of the polyelectrolyte chains throughout our tested PE charge regime scales with the degree of polymerization, following a Flory exponent ($\nu \approx 0.7$) consistent with the behavior of an expanded coil state. This qualitative disagreement with the experimental observations possibly originates from the absence of polar solvents and counterions in our model. Our structural analysis employing Voronoi tessellation analysis reveals that the compactness of the dense coacervate phase increases with the PE chain length and is fairly independent of the effective size (R_g) of the individual constituent PEs, especially for longer-chain PEs, which agrees with existing experimental observation. Longer-chain polyelectrolyte coacervates exhibit greater thermodynamic stability, characterized by increased density, compactness, and higher critical temperature. These long-chain polyelectrolyte chains form more enthalpically favorable multiple contacts, resulting in denser, more compact coacervates that are less susceptible to disruption by thermal fluctuations. Our simulations also report a possible

correlation between the distribution of available space in the PE matrix (Voronoi cavity radii) and the density of PE beads within the coacervate phase, which itself is governed by the degree of polymerization of the constituent chains. We observe that the mean cavity radius scales with polyelectrolyte density, with an intuitive scaling exponent of approximately $-1/3$. In addition, the width of the cavity radius distribution scales with polyelectrolyte density, with a scaling exponent of ~ 0.50 for all values of N tested, although the origin of this scaling factor remains unclear. Given that the present simulation results align well with the experimental trends, we propose that our simplified model of polyelectrolytes, incorporating excluded non-polar solvents and tunable polyelectrolyte interactions, provides a novel framework for investigating the different complex phenomena associated with polyelectrolyte complexation. We note that there are numerous factors known to be influential in the formation of PECs that were not considered here, such as solvent polarity and architectural characteristics, including the shape, symmetry, and topology of polyelectrolytes. We aim to explore these effects in future studies to provide a more comprehensive understanding of polyelectrolyte complexation.

SUPPLEMENTARY MATERIAL

The [supplementary material](#) includes equilibrium protocol details, snapshots of coacervates for varying charges of the PE beads, and tables containing $\langle R_g \rangle$ and $\langle r_{\text{cavity}} \rangle$ values for all simulated systems and corresponding scaling parameter values.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Kayley Alonso and Atanu Baksi contributed equally to this work.

Kayley Alonso: Conceptualization (equal); Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Atanu Baksi:** Conceptualization (equal);

Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Isabel Knight:** Conceptualization (supporting); Data curation (supporting); Investigation (supporting); Writing – original draft (supporting). **Nga Nguyen:** Conceptualization (supporting); Data curation (supporting); Investigation (supporting); Writing – original draft (supporting). **Mohsen Farshad:** Data curation (supporting); Investigation (supporting); Supervision (supporting); Writing – original draft (equal); Writing – review & editing (equal). **Samanvaya Srivastava:** Conceptualization (equal); Formal analysis (equal); Writing – original draft (equal); Writing – review & editing (equal). **Jonathan K. Whitmer:** Conceptualization (lead); Funding acquisition (lead); Investigation (supporting); Methodology (lead); Project administration (lead); Supervision (lead); Writing – original draft (equal); Writing – review & editing (equal).

DATA AVAILABILITY

Data supporting the findings of this study are available in the article and its [supplementary material](#). Codes used in this work are available from the authors upon reasonable request.

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